Class19

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Pertussis and the CMI-PB project

1. Investigating pertussis cases by year

```
library(datapasta)
cdc <- data.frame(
                          Year = c(1922L,
                                   1923L,1924L,1925L,1926L,1927L,1928L,
                                   1929L,1930L,1931L,1932L,1933L,1934L,1935L,
                                   1936L,1937L,1938L,1939L,1940L,1941L,
                                   1942L,1943L,1944L,1945L,1946L,1947L,1948L,
                                   1949L,1950L,1951L,1952L,1953L,1954L,
                                   1955L,1956L,1957L,1958L,1959L,1960L,
                                   1961L,1962L,1963L,1964L,1965L,1966L,1967L,
                                   1968L,1969L,1970L,1971L,1972L,1973L,
                                   1974L,1975L,1976L,1977L,1978L,1979L,1980L,
                                   1981L,1982L,1983L,1984L,1985L,1986L,
                                   1987L,1988L,1989L,1990L,1991L,1992L,1993L,
                                   1994L,1995L,1996L,1997L,1998L,1999L,
                                   2000L,2001L,2002L,2003L,2004L,2005L,
                                   2006L,2007L,2008L,2009L,2010L,2011L,2012L,
                                   2013L,2014L,2015L,2016L,2017L,2018L,
                                   2019L,2020L,2021L,2022L,2023L),
  No..Reported.Pertussis.Cases = c(107473,
                                   164191,165418,152003,202210,181411,
                                   161799,197371,166914,172559,215343,179135,
                                   265269,180518,147237,214652,227319,103188,
                                   183866,222202,191383,191890,109873,
                                   133792,109860,156517,74715,69479,120718,
                                   68687,45030,37129,60886,62786,31732,28295,
                                   32148,40005,14809,11468,17749,17135,
```

```
                                   13005,6799,7717,9718,4810,3285,4249,
                                   3036,3287,1759,2402,1738,1010,2177,2063,
                                   1623,1730,1248,1895,2463,2276,3589,
                                   4195,2823,3450,4157,4570,2719,4083,6586,
                                   4617,5137,7796,6564,7405,7298,7867,
                                   7580,9771,11647,25827,25616,15632,10454,
                                   13278,16858,27550,18719,48277,28639,
                                   32971,20762,17972,18975,15609,18617,6124,
                                   2116,3044,7063)
)
```

```
names(cdc)
```

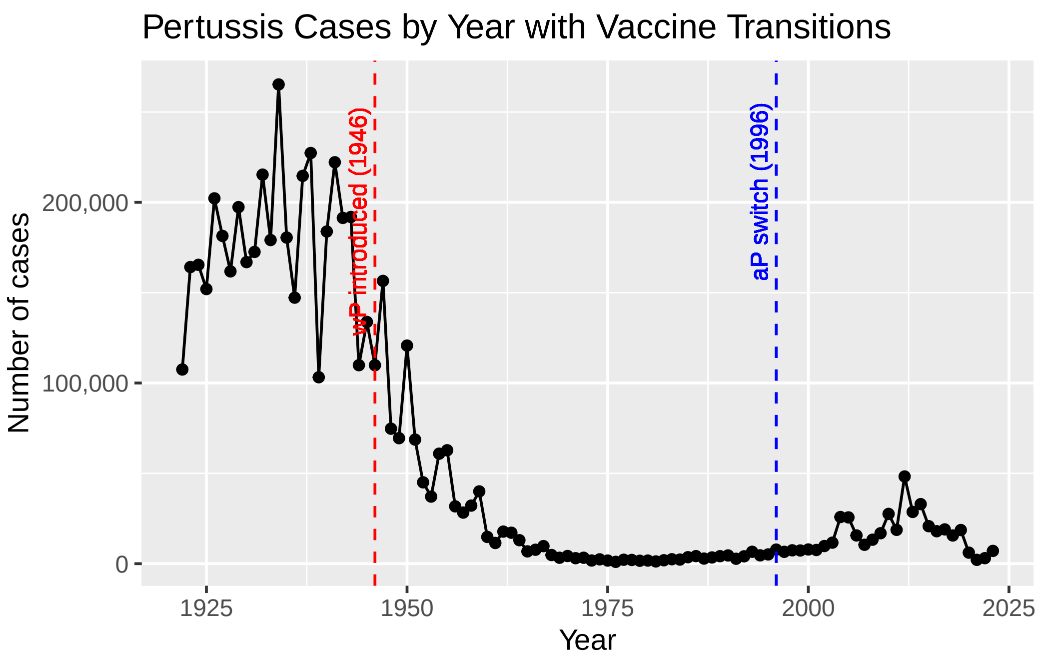
```
[1] "Year"                                "No..Reported.Pertussis.Cases"
```

Q1 and Q2

```
library(ggplot2)
library(scales)

ggplot(cdc) +
  aes(x = Year, y = No..Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Pertussis Cases by Year",
    x = "Year",
    y = "Number of cases"
  ) +   geom_vline(xintercept = 1946, color = "red", linetype = "dashed") +
  geom_vline(xintercept = 1996, color = "blue", linetype = "dashed") +  geom_text(
    aes(x = 1946, y = max(No..Reported.Pertussis.Cases), label = "wP introduced (1946)"),
    color = "red",
    angle = 90,
    vjust = -0.5,
    hjust = 1.1,
    size = 3
  ) +
  geom_text(
    aes(x = 1996, y = max(No..Reported.Pertussis.Cases), label = "aP switch (1996)"),
    color = "blue",
    angle = 90,
```

```
    vjust = -0.5,
    hjust = 1.1,
    size = 3
  ) +

  labs(
    title = "Pertussis Cases by Year with Vaccine Transitions",
    x = "Year",
    y = "Number of cases"
  )
```

```
Warning in geom_text(aes(x = 1946, y = max(No..Reported.Pertussis.Cases), : All aesthetics ha
i Please consider using `annotate()` or provide this layer with data containing
  a single row.

Warning in geom_text(aes(x = 1996, y = max(No..Reported.Pertussis.Cases), : All aesthetics ha
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```



2. A tale of two vaccines (wP & aP)

> Q3. What happened after the introduction of the aP vaccine, and why?

After the switch to the acellular pertussis (aP) vaccine in 1996, pertussis cases in the U.S. began to increase again after decades of very low incidence. By 2012, reported cases reached their highest level since the 1950s. It might be immunity from the aP vaccine wanes more quickly than immunity from the older whole-cell (wP) vaccine. Additional contributing factors may include more sensitive PCR testing, bacterial evolution, and vaccine hesitancy, all of which can increase the number of reported cases.

```r
library(jsonlite)
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject, 3)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
```

> Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```r
table(subject$infancy_vac)
```

```

aP wP 
87 85 
```

There are 60 aP-vaccinated subjects and 58 wP-vaccinated subjects in the dataset.

```r
names(subject)
```

```
[1] "subject_id"     "infancy_vac"    "biological_sex" "ethnicity"     
[5] "race"           "year_of_birth"  "date_of_boost"  "dataset"       
```

> Q5. How many Male and Female subjects?

```r
table(subject$biological_sex)
```

```

Female   Male 
   112     60 
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc…)?

```r
table(subject$biological_sex, subject$race)
```

```

        
         American Indian/Alaska Native Asian Black or African American
  Female                             0    32                         2
  Male                               1    12                         3
        
         More Than One Race Native Hawaiian or Other Pacific Islander
  Female                 15                                         1
  Male                    4                                         1
        
         Unknown or Not Reported White
  Female                      14    48
  Male                         7    32
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```r
library(lubridate)
```

```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

```r
today()
```

```
[1] "2025-12-03"
```

```r
subject$age <- today() - ymd(subject$year_of_birth)
```

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
ap <- subject %>% filter(infancy_vac == "aP")

round( summary( time_length( ap$age, "years" ) ) )
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
     23      27      28      28      29      35 
```

```r
wp <- subject %>% filter(infancy_vac == "wP")

round(summary(time_length(wp$age, "years")))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
     23      33      35      37      40      58 
```

Q8. Determine the age of all individuals at time of boost?

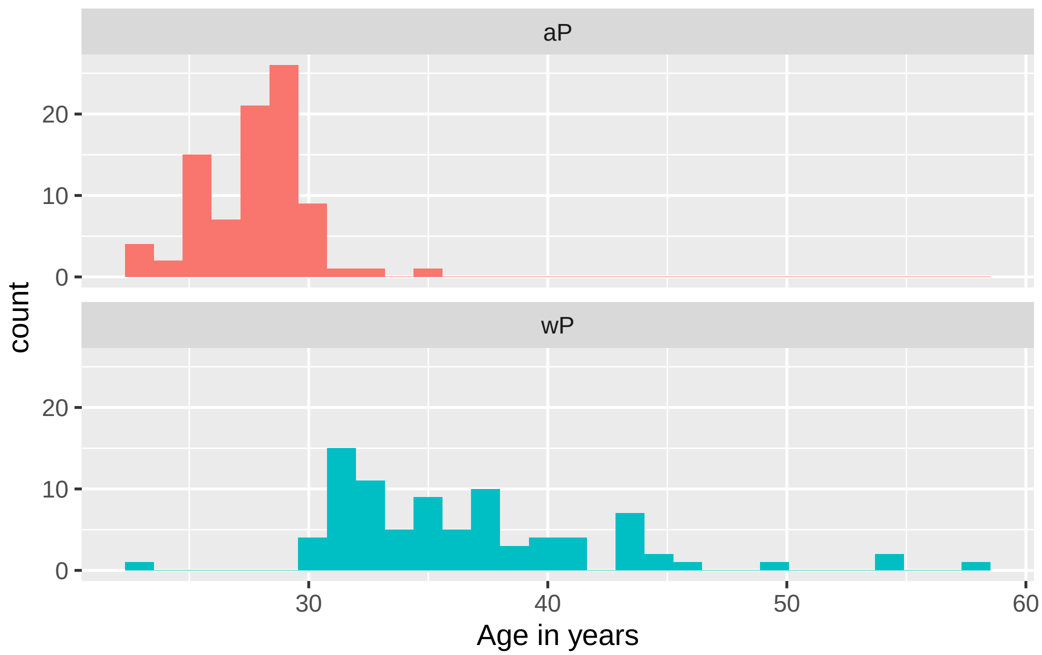
```r
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```r
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```r
# Or wilcox.test()
x <- t.test(time_length( wp$age, "years" ),
       time_length( ap$age, "years" ))

x$p.value
```

```
[1] 2.372101e-23
```

Joining multiple table

```r
library(jsonlite)
specimen <- read_json("https://www.cmi-pb.org/api/v4/specimen", simplifyVector = TRUE)
titer    <- read_json("https://www.cmi-pb.org/api/v4/plasma_ab_titer", simplifyVector = TRUE)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```r
meta <- left_join(specimen, subject, by = "subject_id")
dim(meta)
```

```
[1] 939  14
```

```r
head(meta)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood     1          wP         Female
2                             1         Blood     2          wP         Female
3                             3         Blood     3          wP         Female
4                             7         Blood     4          wP         Female
5                            14         Blood     5          wP         Female
6                            30         Blood     6          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
         age
1 14581 days
2 14581 days
3 14581 days
```

```
4 14581 days
5 14581 days
6 14581 days
```

> Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```r
abdata <- inner_join(titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
dim(abdata)
```

```
[1] 41775    21
```

> Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```r
table(abdata$isotype)
```

```

 IgE  IgG IgG1 IgG2 IgG3 IgG4 
6698 3233 7961 7961 7961 7961 
```

> Q12. What are the different dataset values in abdata, and what do you notice?

```r
table(abdata$dataset)
```

```

2020_dataset 2021_dataset 2022_dataset 
       31520         8085         2170 
```

4. Examine IgG Ab titer levels

```r
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

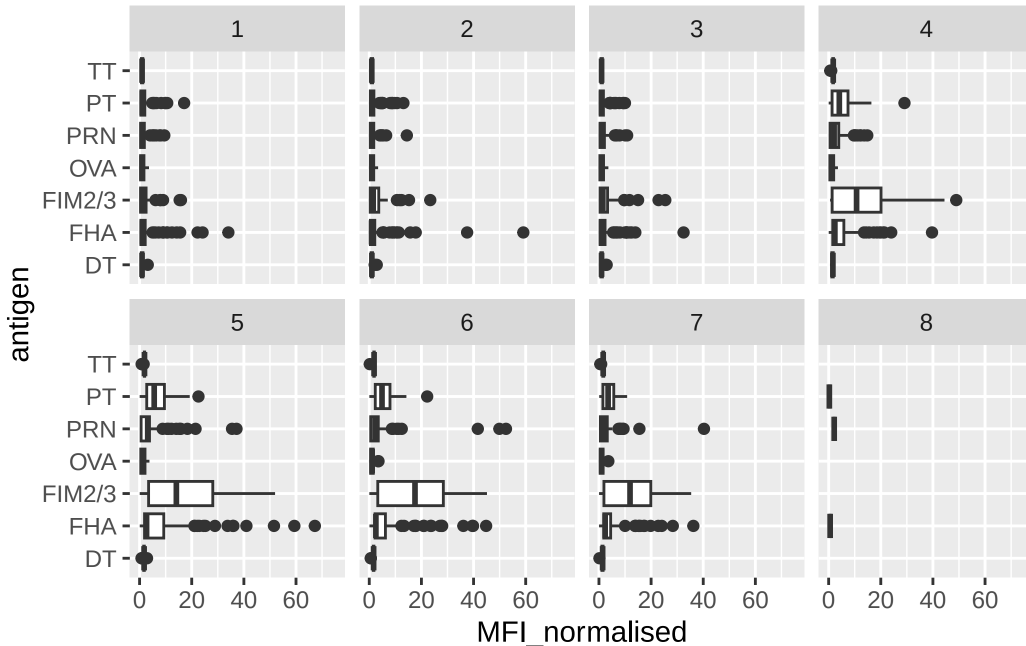
```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id actual_day_relative_to_boost
1 IU/ML                 0.530000          1                           -3
2 IU/ML                 6.205949          1                           -3
3 IU/ML                 4.679535          1                           -3
4 IU/ML                 0.530000          3                           -3
5 IU/ML                 6.205949          3                           -3
6 IU/ML                 4.679535          3                           -3
  planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood     1          wP         Female
2                             0         Blood     1          wP         Female
3                             0         Blood     1          wP         Female
4                             0         Blood     1          wP         Female
5                             0         Blood     1          wP         Female
6                             0         Blood     1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
         age
1 14581 days
2 14581 days
3 14581 days
4 15677 days
5 15677 days
6 15677 days
```

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(x = MFI_normalised, y = antigen) +
  geom_boxplot() +
```

```r
  xlim(0, 75) +
  facet_wrap(vars(visit), nrow = 2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

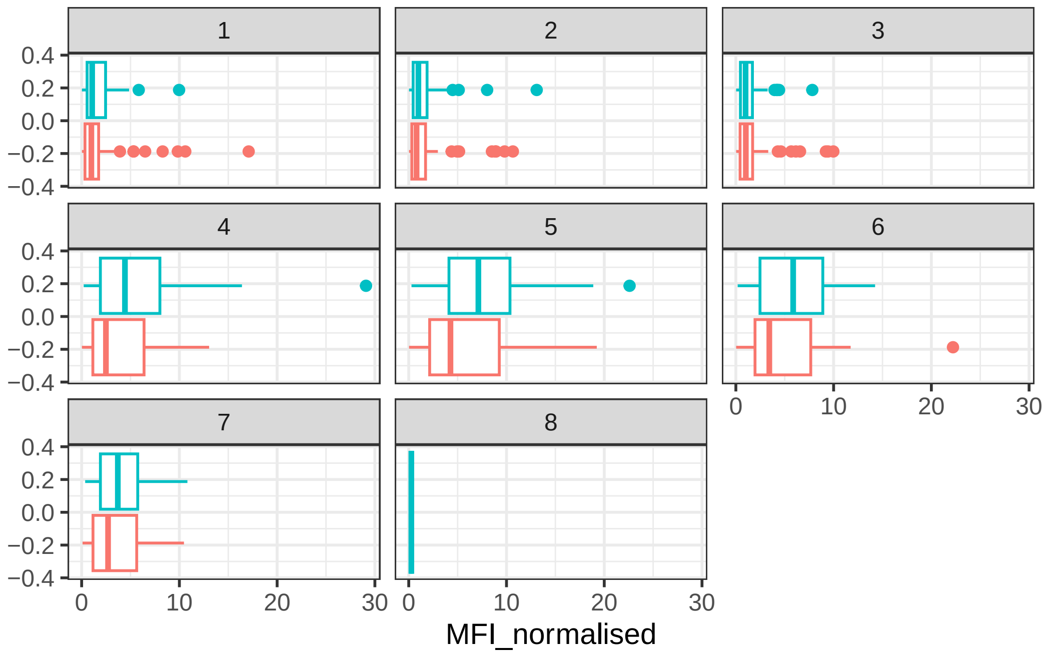


> Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

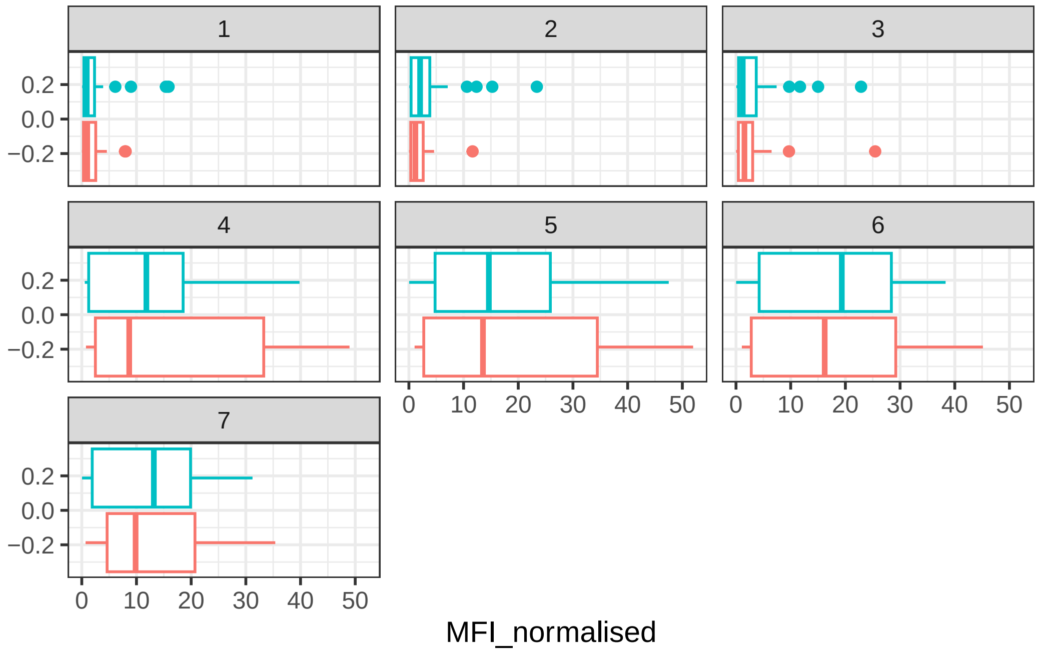
Pertussis-specific antigens (PT, PRN, FHA, FIM2/3) show clear changes in IgG titers over time because they are components of the pertussis booster vaccine and elicit a vaccine-driven response. Non-pertussis antigens (TT, DT, OVA) show little change because the subjects were not recently vaccinated against them or they are negative controls.

> Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a "control" antigen ("OVA", that is not in our vaccines) and a clear antigen of interest ("PT", Pertussis Toxin, one of the key virulence factors produced by the bacterium B. pertussis).

```r
filter(igg, antigen == "PT") %>%
  ggplot() +
  aes(x = MFI_normalised, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```




```r
filter(igg, antigen == "FIM2/3") %>%
  ggplot() +
  aes(x = MFI_normalised, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

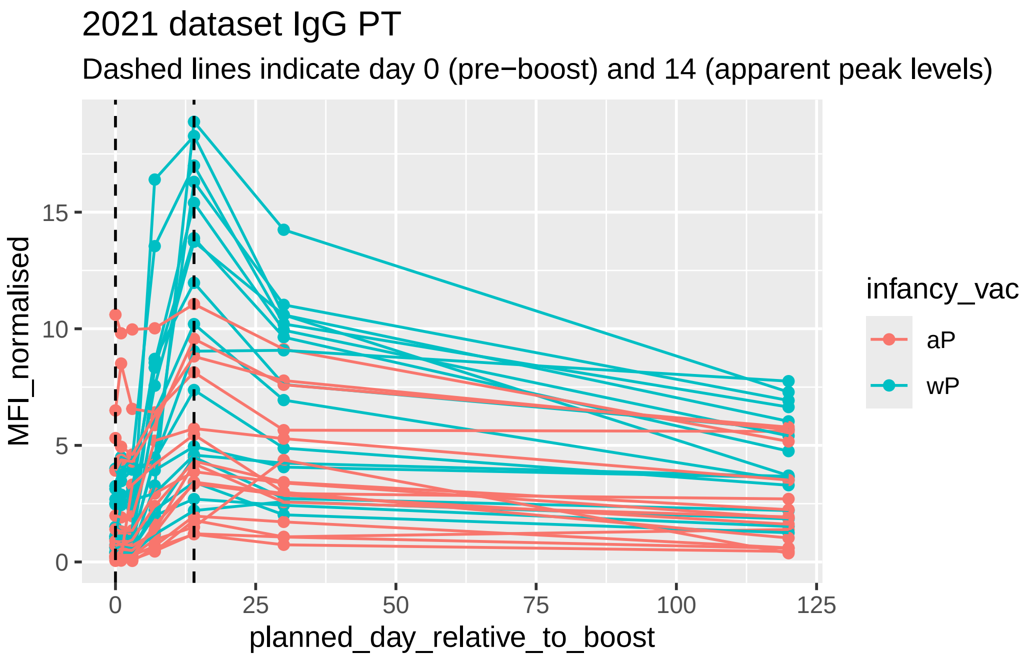
PT levels clearly rise after boosting, peak around visit 5–6, and then decline. OVA does not change over time because it is not a vaccine antigen. The PT pattern looks similar for both aP and wP subjects.

Q17. Do you see any clear difference in aP vs. wP responses?

For OVA, aP and wP responses look the same — both groups show low, flat background levels across visits, with no boosting. There is no meaningful difference between the groups.

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG",  antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
```

```
  geom_vline(xintercept=14, linetype="dashed") +
labs(title="2021 dataset IgG PT",
     subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



Q18. Does this trend look similar for the 2020 dataset?

Yes. The 2020 dataset shows the same basic pattern seen in 2021: PT IgG levels rise sharply after boosting, reach a peak near day 14, and decline thereafter. Both aP and wP subjects follow this same time course, with no major differences between groups.